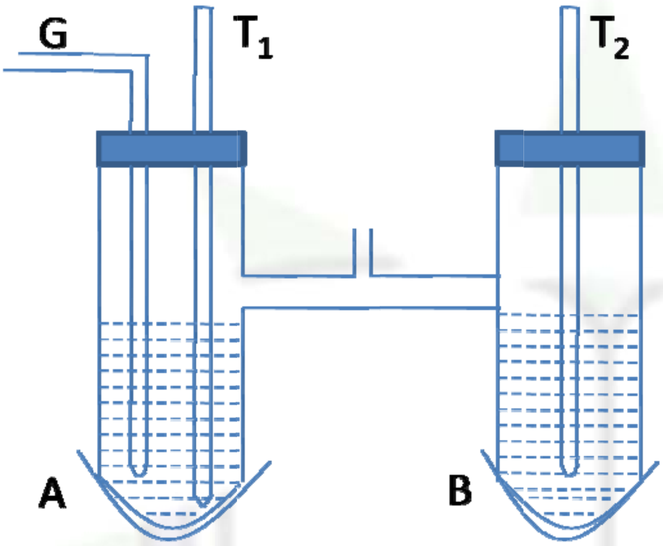
Determination of relative humidity by Renault's hygrometer:

Theory :

$$\text{R.H} = \frac{\text{The S.V.P at the corresponding due point}}{\text{The S.V.P at } t^0\text{C}} \times 100$$



Description: Renault's hygrometer consists of two test tube A and B containing ether as shown. Two silver tumbles S are fitted at the bottom of the test tubes. Two thermometers T_1 and T_2 are inserted in the two test tubes A and B respectively. Test tube A is provided with a delivery tube G and is connected to an aspirator as shown. When aspirator is operated air is drawn through ether by the delivery tube and ether evaporates. Ether requires latent heat for evaporation which is absorbed from the liquid itself and hence temperature of the test tube A falls.

The temperature soon reaches the dew point which is indicated by a cloudy appearance of the silver tumble S in A. This becomes evident when we compare it with the silver tumble S fitted in test tube B, which remains shining as usual. The temperature at which this occurs is noted from the thermometer T_1, aspirator is then closed no evaporation of ether takes place and the temperature of the test tube A rises and the temperature at which the dews clears is noted again from thermometer T_1 , the mean of the two readings gives the dew point. The atmospheric temperature is recorded from the thermometer T_2. From the vapor pressure table S.V.P at due point and S.V.P at atm. temperature are noted and hence the RH can be calculated.